

Midterm Mock Test Solutions

MTH253 – Calculus III

1 Key Formulas

Definition 1 (Directional Derivative). The *directional derivative* of $f(x, y)$ in the direction of a **unit vector** $\vec{u} = \langle a, b \rangle$ is

$$D_{\vec{u}} f(x, y) = f_x(x, y) a + f_y(x, y) b = \nabla f(x, y) \cdot \vec{u}.$$

Two common situations:

- (i) **Direction given by an angle θ :** Set $\vec{u} = \langle \cos \theta, \sin \theta \rangle$, so

$$D_{\vec{u}} f(x, y) = f_x(x, y) \cos \theta + f_y(x, y) \sin \theta.$$

- (ii) **Direction given by a (non-unit) vector $\vec{v}$:** You **must** first normalise:

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|},$$

then apply the formula above.

2 General Solution Method

Given $f(x, y)$, a point $P = (x_0, y_0)$, and a direction (angle or vector):

1. **Compute the partial derivatives** f_x and f_y .
2. **Evaluate at the point:** $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$.
3. **Form the gradient:** $\nabla f(x_0, y_0) = \langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle$.
4. **Obtain the unit direction vector $\vec{u}$:**
 - If an angle θ is given: $\vec{u} = \langle \cos \theta, \sin \theta \rangle$.
 - If a vector $\vec{v}$ is given: $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$.
5. **Dot product:** $D_{\vec{u}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \vec{u}$.

3 Worked Example 1 – Direction Given by an Angle

Example 1. Find the directional derivative of $f(x, y) = x^2y^3 - 4y$ at the point $(2, -1)$ in the direction $\theta = \pi/6$.

Solution.

Step 1. Partial derivatives:

$$f_x = 2xy^3, \quad f_y = 3x^2y^2 - 4.$$

Step 2. Evaluate at $(2, -1)$:

$$f_x(2, -1) = 2(2)(-1)^3 = -4, \quad f_y(2, -1) = 3(4)(1) - 4 = 8.$$

Step 3. Gradient:

$$\nabla f(2, -1) = \langle -4, 8 \rangle.$$

Step 4. Unit vector from angle $\theta = \pi/6$:

$$\vec{u} = \left\langle \cos \frac{\pi}{6}, \sin \frac{\pi}{6} \right\rangle = \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle.$$

Step 5. Directional derivative:

$$D_{\vec{u}} f(2, -1) = \langle -4, 8 \rangle \cdot \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle = -4 \cdot \frac{\sqrt{3}}{2} + 8 \cdot \frac{1}{2} = -2\sqrt{3} + 4.$$

$$D_{\vec{u}} f(2, -1) = 4 - 2\sqrt{3} \approx 0.536$$

4 Worked Example 2 – Direction Given by a Vector

Example 2. Find the directional derivative of $f(x, y) = e^x \sin y$ at $(0, \pi/3)$ in the direction of $\vec{v} = \langle 3, 4 \rangle$.

Solution.

Step 1. Partial derivatives:

$$f_x = e^x \sin y, \quad f_y = e^x \cos y.$$

Step 2. Evaluate at $(0, \pi/3)$:

$$f_x(0, \frac{\pi}{3}) = e^0 \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \quad f_y(0, \frac{\pi}{3}) = e^0 \cos \frac{\pi}{3} = \frac{1}{2}.$$

Step 3. Gradient:

$$\nabla f(0, \frac{\pi}{3}) = \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle.$$

Step 4. Normalise $\vec{v} = \langle 3, 4 \rangle$:

$$\|\vec{v}\| = \sqrt{9 + 16} = 5, \quad \vec{u} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle.$$

Step 5. Directional derivative:

$$D_{\vec{u}} f\left(0, \frac{\pi}{3}\right) = \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle \cdot \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle = \frac{3\sqrt{3}}{10} + \frac{4}{10} = \frac{3\sqrt{3} + 4}{10}.$$

$$\boxed{D_{\vec{u}} f\left(0, \frac{\pi}{3}\right) = \frac{3\sqrt{3} + 4}{10} \approx 0.920}$$

5 Question 3 – Extreme Values on a Closed Disk

Example 3. Find the extreme values of the function

$$f(x, y) = x^2 + y^2 + 4x - 4y$$

on the disk $x^2 + y^2 \leq 9$.

Solution.

Since the disk $D = \{(x, y) : x^2 + y^2 \leq 9\}$ is a **closed and bounded** region, by the Extreme Value Theorem f attains both an absolute maximum and an absolute minimum on D . We must check:

- (a) Interior critical points (where $\nabla f = \vec{0}$).
- (b) The boundary $x^2 + y^2 = 9$.

Part (a): Interior critical points

Compute the partial derivatives and set them to zero:

$$f_x = 2x + 4 = 0 \implies x = -2, \quad f_y = 2y - 4 = 0 \implies y = 2.$$

The only critical point is $(-2, 2)$. Check whether it lies inside the disk:

$$(-2)^2 + 2^2 = 4 + 4 = 8 \leq 9 \quad \checkmark$$

So $(-2, 2)$ is in the disk. Evaluate f there:

$$f(-2, 2) = 4 + 4 + 4(-2) - 4(2) = 8 - 8 - 8 = -8.$$

Part (b): Boundary $x^2 + y^2 = 9$

Parametrise the circle of radius 3:

$$x = 3 \cos \theta, \quad y = 3 \sin \theta, \quad 0 \leq \theta < 2\pi.$$

Substitute into f :

$$\begin{aligned} g(\theta) &= (3 \cos \theta)^2 + (3 \sin \theta)^2 + 4(3 \cos \theta) - 4(3 \sin \theta) \\ &= 9 \cos^2 \theta + 9 \sin^2 \theta + 12 \cos \theta - 12 \sin \theta \\ &= 9 + 12 \cos \theta - 12 \sin \theta. \end{aligned}$$

To find extrema of $g(\theta)$, set $g'(\theta) = 0$:

$$g'(\theta) = -12 \sin \theta - 12 \cos \theta = 0 \implies \sin \theta = -\cos \theta \implies \tan \theta = -1.$$

This gives two solutions in $[0, 2\pi)$:

$$\theta_1 = \frac{3\pi}{4}, \quad \theta_2 = \frac{7\pi}{4}.$$

Evaluate g at each:

At $\theta_1 = \frac{3\pi}{4}$:

$$\cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}.$$

$$g\left(\frac{3\pi}{4}\right) = 9 + 12\left(-\frac{\sqrt{2}}{2}\right) - 12\left(\frac{\sqrt{2}}{2}\right) = 9 - 6\sqrt{2} - 6\sqrt{2} = 9 - 12\sqrt{2}.$$

At $\theta_2 = \frac{7\pi}{4}$:

$$\cos \frac{7\pi}{4} = \frac{\sqrt{2}}{2}, \quad \sin \frac{7\pi}{4} = -\frac{\sqrt{2}}{2}.$$

$$g\left(\frac{7\pi}{4}\right) = 9 + 12\left(\frac{\sqrt{2}}{2}\right) - 12\left(-\frac{\sqrt{2}}{2}\right) = 9 + 6\sqrt{2} + 6\sqrt{2} = 9 + 12\sqrt{2}.$$

The corresponding points on the boundary are:

$$\theta_1 : \left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right), \quad \theta_2 : \left(\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}\right).$$

Conclusion

Compare all candidate values:

Point	Type	$f(x, y)$
$(-2, 2)$	Interior critical pt.	-8
$\left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$	Boundary	$9 - 12\sqrt{2} \approx -7.97$
$\left(\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}\right)$	Boundary	$9 + 12\sqrt{2} \approx 25.97$

Absolute minimum = -8 at $(-2, 2)$

Absolute maximum = $9 + 12\sqrt{2}$ at $\left(\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}\right)$

6 Question 4 – Volume via Double Integral

Example 4. Find the volume of the solid under the surface

$$z = 1 + x^2y^2$$

and above the region enclosed by $x = y^2$ and $x = 4$.

Solution.

The volume is given by the double integral

$$V = \iint_R (1 + x^2y^2) dA,$$

where R is the region in the xy -plane bounded by the parabola $x = y^2$ and the vertical line $x = 4$.

Step 1: Sketch the region and determine limits

The curves $x = y^2$ and $x = 4$ intersect when

$$y^2 = 4 \implies y = \pm 2.$$

For a fixed $y \in [-2, 2]$, x runs from the parabola to the line:

$$y^2 \leq x \leq 4.$$

So the integral becomes

$$V = \int_{-2}^2 \int_{y^2}^4 (1 + x^2y^2) dx dy.$$

Step 2: Evaluate the inner integral (with respect to x)

$$\begin{aligned} \int_{y^2}^4 (1 + x^2y^2) dx &= \left[x + \frac{x^3}{3} y^2 \right]_{x=y^2}^{x=4} \\ &= \left(4 + \frac{64}{3} y^2 \right) - \left(y^2 + \frac{y^6}{3} y^2 \right) \\ &= 4 + \frac{64}{3} y^2 - y^2 - \frac{y^8}{3} \\ &= 4 + \frac{64y^2 - 3y^2}{3} - \frac{y^8}{3} \\ &= 4 + \frac{61y^2}{3} - \frac{y^8}{3}. \end{aligned}$$

Step 3: Evaluate the outer integral (with respect to y)

Since the integrand $4 + \frac{61y^2}{3} - \frac{y^8}{3}$ is an **even function** of y (only even powers), we can use symmetry:

$$V = 2 \int_0^2 \left(4 + \frac{61y^2}{3} - \frac{y^8}{3} \right) dy.$$

Evaluate each term:

$$\int_0^2 4 \, dy = 4 \cdot 2 = 8,$$

$$\int_0^2 \frac{61y^2}{3} \, dy = \frac{61}{3} \cdot \frac{y^3}{3} \Big|_0^2 = \frac{61}{3} \cdot \frac{8}{3} = \frac{488}{9},$$

$$\int_0^2 \frac{y^8}{3} \, dy = \frac{1}{3} \cdot \frac{y^9}{9} \Big|_0^2 = \frac{1}{3} \cdot \frac{512}{9} = \frac{512}{27}.$$

Combine:

$$\begin{aligned} \frac{V}{2} &= 8 + \frac{488}{9} - \frac{512}{27} \\ &= \frac{216}{27} + \frac{1464}{27} - \frac{512}{27} \quad (\text{common denominator } 27) \\ &= \frac{216 + 1464 - 512}{27} \\ &= \frac{1168}{27}. \end{aligned}$$

Therefore:

$$V = 2 \cdot \frac{1168}{27} = \frac{2336}{27}.$$

$$\boxed{V = \frac{2336}{27} \approx 86.52}$$